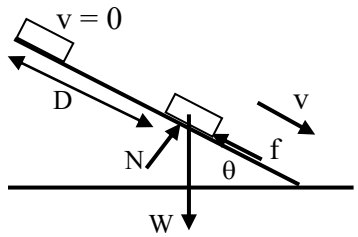
Group Problem 6: Solution

You are investigating the use of rescue sleds pulled by snowmobiles for snow emergencies. To determine how this could work, you first need to measure the coefficient of friction of a typical snowpack on a moving fully loaded 160 kg sled. To accomplish this, you take a fully loaded sled up a hill, which is at 15° to the horizontal and covered with a typical snowpack. You then release the sled so that it slides down the hill. Borrowing a police radar gun, you determine that the sled goes 10 miles/hour and is speeding up when it is 120 feet from its release point.

Listed below is an example of 20 decisions for one organized path of solving this problem. Your organized solution path may differ. Now is a good time to write down those mental steps to reinforce your process of organized decisions to solve a problem.

1. To begin the decision-making process, draw a useful sketch. That sketch should have your best guess about the important objects in the problem, their interactions, and their motion.
2. In this case, a sled goes down a hill with its speed changing from 0 at its start to a known speed after a known distance. This information should go on the picture.
3. Since its speed is changing, the sled is accelerating so it is likely the object of interest. Make sure the sled's velocity at the beginning and end of the time interval are on the sketch.
4. Put all the sled's interactions (forces on the sled) on the sketch.
5. Decide on what the question is asking and what that question has to do with the motion and interactions of the sled. In this case the coefficient of friction, which must be kinetic friction since the sled is moving, is the target of the problem.
6. The coefficient of kinetic friction determines the frictional force on the sled. That force also depends on the normal force on the sled. Make sure both the frictional force and the normal force on the sled are on the sketch. Determine what other forces are acting on the sled and put them on the sketch. Because this is on the surface of the Earth, the gravitational force of the Earth on the sled (the sled's weight) is always there.
7. Put everything that is known about the situation that might affect the sled's motion on the picture.



Sled sliding downhill

Know
$\theta = 15^o$
m = 160 kg
v = 10 mph
D = 120 ft

Question: What is the coefficient of kinetic friction?
Target: μ_k

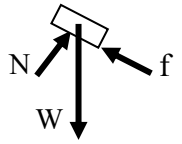
8. Decide any physics equations that could be useful to determine the answer to your question.

$f = \mu_k N$ where μ_k is the coefficient of kinetic friction. Perhaps W = mg.

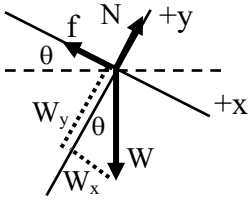
9. To find μ_k, need to determine two forces on the sled, the frictional force (f) and the normal force (N).
10. Because of the need to determine forces on the sled, isolate the sled and draw the forces on it. (a free body diagram).
11. Since the forces are in different directions, the perpendicular components of the forces need to be treated independently. To do this, put the forces on a coordinate system (force diagram).

12. Choose a coordinate system that will have the fewest components to make the algebra easier. A system with one axis along the hill and the other perpendicular to the hill has only one force whose components need to be found, the weight.
13. Use $\sum \vec{F} = m\vec{a}$ to write down equations along each axis relating the forces.

Free Body Diagram
Sled downhill



Force Diagram of
Sled downhill



$$\sum F_x : W_x - f = ma_x$$

$$\sum F_y : 0 = N - W_y$$

$$\sin\theta = \frac{W_x}{W} \qquad W_x = W\sin\theta$$

$$\cos\theta = \frac{W_y}{W} \qquad W_y = W\cos\theta$$

$$0 = N - W\cos\theta$$

$$\boxed{W\cos\theta = N}$$

Since W and θ are known, N is determined.

14. All that is left is to determine f. You cannot easily use $\sum F_x$ because it is not 0 but equal to ma and you don't know a (the acceleration of the sled).
15. Looking back at the sketch, you know that the sled moves down the hill, its starting velocity and its final velocity after going a distance D. This velocity change might be determined by conservation of energy.
16. In this case, the velocity change happens over a known distance. Conservation of energy relates the change of the energy of an object to energy transfer caused by a force on the object exerted over a distance. Choose conservation of energy to find the frictional force on the sled.
17. Choose the sled as the system of interest in which case the frictional force causes and energy output from the sled and the component of weight along the hill causes an energy input to the sled. Since this system only has one object, the only energy it can have is kinetic energy neglecting its internal structure.

Use conservation of energy to determine f.
System: the sled
Initial time: just after the sled is released on the hill.
Final time: when the sled goes a distance D.

$$\Delta E_{sled} = \Delta E_{transfer}$$

$$KE_{sled\,f} - KE_{sled\,i} = E_{input} - E_{output}$$

$$\frac{1}{2}mv^2 - 0 = W_x D - fD$$

$$\frac{1}{2}mv^2 = mg\sin\theta\, D - fD$$

The only unknown is f

$$fD = mg\sin\theta\, D - \frac{1}{2}mv^2$$

$$\boxed{f = mg\sin\theta - \frac{1}{2D}mv^2}$$

Determine μ_k from $f = \mu_k N$ and $W\cos\theta = N$

$$\left(mg\sin\theta - \frac{1}{2D}mv^2\right) = \mu_k(mg\cos\theta)$$

$$g\sin\theta - \frac{1}{2D}v^2 = \mu_k g\cos\theta$$

$$\frac{\sin\theta}{\cos\theta} - \frac{1}{2Dg\cos\theta}v^2 = \mu_k$$

$$\boxed{\tan\theta - \frac{1}{2Dg\cos\theta}v^2 = \mu_k}$$

18. Before putting in numbers, check to see if you have made a mistake. That way you don't waste a time with the numbers. First check the units because that is easiest and then check if the behavior of the answer equation is reasonable.

Evaluate the answer equation

Check units

$$[\tan\theta] - \frac{1}{2[D][g][\cos\theta]}[v^2] = [\mu_k]$$

$$[1] - \frac{1}{[m]\left[\frac{m}{s^2}\right]}\left[\frac{m^2}{s^2}\right] = [\mu_k]$$

$$\frac{\left[\frac{m^2}{s^2}\right]}{\left[\frac{m^2}{s^2}\right]} = [\mu_k]$$

$$[1] = [\mu_k]$$

The units check since μ_k has no units.

Is the behavior of the answer equation reasonable?

If v gets larger, μ_k would have to be smaller because the v^2 term is subtracted. This is reasonable since a larger final velocity means the frictional force needs to be smaller (less energy output). A smaller frictional force requires a smaller μ_k.

If D gets larger, μ_k would have to be larger. This is reasonable since needing a larger distance (more energy input from the gravitational force) to get the same speed means the frictional force is larger (more energy output).

19. Since the answer equation has the correct units and a reasonable behavior, it is not obviously wrong. Now put in numbers. Some unit conversion is needed to go from miles per hour to feet per second. Do the unit conversion last because sometimes the pesky units cancel out.

Put in numbers:

$$\tan 15° - \frac{1}{2(120\text{ ft})\left(32\frac{\text{ft}}{\text{s}^2}\right)\cos 15°}\left(10\frac{\text{mi}}{\text{hr}}\right)^2 = \mu_k$$

$$0.27 - \frac{100}{7680\left(\frac{\text{ft}^2}{\text{s}^2}\right)0.97}\left(\frac{\text{mi}}{\text{hr}}\right)^2 = \mu_k$$

$$0.27 - \frac{100}{7450\left(\frac{\text{ft}^2}{\text{s}^2}\right)}\left(\frac{5280\text{ ft}}{1\text{ mile}}\right)^2\left(\frac{1\text{ hr}}{3600\text{ s}}\right)^2\left(\frac{\text{mi}}{\text{hr}}\right)^2 = \mu_k$$

$$0.27 - 0.03 = \mu_k$$

$$\boxed{0.24 = \mu_k}$$

20. Check that there are no obvious mistakes made when putting in the numbers by checking the units and the reasonableness of the final numerical value.

The units are correct since μ_k has no units.

0.24 is not an unreasonable value since values of coefficients of kinetic friction are between 0 and 1.